

DE1-SoC Switch Testing Guide

Camera, Video Processing & VGA Display

Switch & Key Map Overview

	System Reset (keep DOWN = 0 during normal operation)
SW[2:1]	Red intensity: 00=Full 01=Half 10=Quarter 11=Off
SW[4:3]	Green intensity: 00=Full 01=Half 10=Quarter 11=Off
SW[6:5]	Blue intensity: 00=Full 01=Half 10=Quarter 11=Off
SW[7]	Cursor select: 0 = move Cursor A 1 = move Cursor B
SW[8]	Grayscale: 0 = colour 1 = grayscale
SW[9]	LED[0] demo (not used for video processing)
KEY[3]	Move selected cursor UP (active-LOW)
KEY[2]	Move selected cursor DOWN (active-LOW)
KEY[1]	Move selected cursor LEFT (active-LOW)
KEY[0]	Move selected cursor RIGHT (active-LOW)

NOTE: SW[0] must always be DOWN (0). Flipping it UP resets the entire system.

PART 1 — RGB Colour Filter Tests

Test 1.1 — Baseline (all filters off)

SW9	SW8	SW7	SW6	SW5	SW4	SW3	SW2	SW1	SW0
DN	DN	DN	DN	DN	DN	DN	DN	DN	DN

Expected:

Normal full-colour camera image.

Corner markers visible: red (top-left), green (top-right), blue (bottom-left).

Test 1.2 — Red Channel

SW[1] UP — Red at HALF

SW9	SW8	SW7	SW6	SW5	SW4	SW3	SW2	SW1	SW0
DN	DN	DN	DN	DN	DN	DN	DN	UP	DN

Expected:

Image looks slightly cool / greenish.

SW[2] UP — Red at QUARTER

SW9	SW8	SW7	SW6	SW5	SW4	SW3	SW2	SW1	SW0
DN	DN	DN	DN	DN	DN	DN	UP	DN	DN

Expected:

Image looks noticeably green/blue tinted.

SW[2:1] UP — Red OFF

SW9	SW8	SW7	SW6	SW5	SW4	SW3	SW2	SW1	SW0
DN	DN	DN	DN	DN	DN	DN	UP	UP	DN

Expected:

No red at all. Skin tones look teal/sick.

Test 1.3 — Green Channel

SW[3] UP — Green at HALF

SW9	SW8	SW7	SW6	SW5	SW4	SW3	SW2	SW1	SW0
DN	DN	DN	DN	DN	DN	UP	DN	DN	DN

Expected:

Image looks pinkish / purple.

SW[4] UP — Green at QUARTER

SW9	SW8	SW7	SW6	SW5	SW4	SW3	SW2	SW1	SW0
DN	DN	DN	DN	DN	UP	DN	DN	DN	DN

Expected:

Image looks strongly magenta.

SW[4:3] UP — Green OFF

SW9	SW8	SW7	SW6	SW5	SW4	SW3	SW2	SW1	SW0
DN	DN	DN	DN	DN	UP	UP	DN	DN	DN

Expected:

No green. Image is red+blue = magenta/purple.

Test 1.4 — Blue Channel

SW[5] UP — Blue at HALF

SW9	SW8	SW7	SW6	SW5	SW4	SW3	SW2	SW1	SW0
DN	DN	DN	DN	UP	DN	DN	DN	DN	DN

Expected:

Image looks slightly warm / yellow.

SW[6] UP — Blue at QUARTER

SW9	SW8	SW7	SW6	SW5	SW4	SW3	SW2	SW1	SW0
DN	DN	DN	UP	DN	DN	DN	DN	DN	DN

Expected:

Image looks warm orange.

SW[6:5] UP — Blue OFF

SW9	SW8	SW7	SW6	SW5	SW4	SW3	SW2	SW1	SW0
DN	DN	DN	UP	UP	DN	DN	DN	DN	DN

Expected:

No blue. Image is red+green = warm yellow/orange. Like an old photograph.

Test 1.5 — Single Colour Only

Red Only (Green OFF + Blue OFF)

SW9	SW8	SW7	SW6	SW5	SW4	SW3	SW2	SW1	SW0
DN	DN	DN	UP	UP	UP	UP	DN	DN	DN

Expected:

Everything in shades of red. Like a darkroom.

Green Only (Red OFF + Blue OFF)

SW9	SW8	SW7	SW6	SW5	SW4	SW3	SW2	SW1	SW0
DN	DN	DN	UP	UP	DN	DN	UP	UP	DN

Expected:

Everything in shades of green. Like night-vision.

Blue Only (Red OFF + Green OFF)

SW9	SW8	SW7	SW6	SW5	SW4	SW3	SW2	SW1	SW0
DN	DN	DN	DN	DN	UP	UP	UP	UP	DN

Expected:

Everything in shades of blue.

Test 1.6 — Two Channels Combined

Yellow (Red + Green, Blue OFF)

SW9	SW8	SW7	SW6	SW5	SW4	SW3	SW2	SW1	SW0
DN	DN	DN	UP	UP	DN	DN	DN	DN	DN

Expected:

Warm yellow/orange image.

Cyan (Green + Blue, Red OFF)

SW9	SW8	SW7	SW6	SW5	SW4	SW3	SW2	SW1	SW0
DN	DN	DN	DN	DN	DN	DN	UP	UP	DN

Expected:

Cool cyan/teal image.

Magenta (Red + Blue, Green OFF)

SW9	SW8	SW7	SW6	SW5	SW4	SW3	SW2	SW1	SW0
DN	DN	DN	DN	DN	UP	UP	DN	DN	DN

Expected:

Purple/magenta image.

Test 1.7 — All Channels Reduced

All at HALF

SW9	SW8	SW7	SW6	SW5	SW4	SW3	SW2	SW1	SW0
DN	DN	DN	DN	UP	DN	UP	DN	UP	DN

Expected:

Same colours but overall darker, like reducing brightness.

All at QUARTER

SW9	SW8	SW7	SW6	SW5	SW4	SW3	SW2	SW1	SW0
DN	DN	DN	UP	DN	UP	DN	UP	DN	DN

Expected:

Very dark image, barely visible.

All OFF

SW9	SW8	SW7	SW6	SW5	SW4	SW3	SW2	SW1	SW0
DN	DN	DN	UP	UP	UP	UP	UP	UP	DN

Expected:

Completely black screen (corner markers still visible).

PART 2 — Cursor Tests

Test 2.1 — Cursor A: 3x3 Solid Block (SW[7]=0)

SW9	SW8	SW7	SW6	SW5	SW4	SW3	SW2	SW1	SW0
DN	DN	DN	DN	DN	DN	DN	DN	DN	DN

Press each KEY button one at a time:

	Cursor moves UP
KEY[2]	Cursor moves DOWN
KEY[1]	Cursor moves LEFT
KEY[0]	Cursor moves RIGHT

Expected:

Tiny 3x3 cyan square moves smoothly (~95 steps/sec).
Starts at position (100, 100) after reset.

Test 2.2 — Cursor A Edge Clamping

SW9	SW8	SW7	SW6	SW5	SW4	SW3	SW2	SW1	SW0
DN	DN	DN	DN	DN	DN	DN	DN	DN	DN

Hold **KEY[0]** (right) for several seconds. Repeat for all four edges.

Expected:

Cursor slides to the edge and stops. It never disappears or wraps around.

Test 2.3 — Cursor B: 10x10 Hollow Box (SW[7]=1)

SW9	SW8	SW7	SW6	SW5	SW4	SW3	SW2	SW1	SW0
DN	DN	UP	DN	DN	DN	DN	DN	DN	DN

Press all four KEY buttons.

Expected:

10x10 hollow cyan square (only border, hollow inside) moves around.
Starts at position (300, 200) after reset.

Test 2.4 — Switching Between Cursors

Step 1: SW[7]=0 — use KEYs to move Cursor A.

SW9	SW8	SW7	SW6	SW5	SW4	SW3	SW2	SW1	SW0
DN	DN	DN	DN	DN	DN	DN	DN	DN	DN

Step 2: Flip SW[7]=1 — now KEYs move Cursor B.

SW9	SW8	SW7	SW6	SW5	SW4	SW3	SW2	SW1	SW0
DN	DN	UP	DN	DN	DN	DN	DN	DN	DN

Expected:

Cursor A freezes in place.

Cursor B starts moving.

Both cursors remain visible simultaneously.

Test 2.5 — Both Cursors Overlapping

Move Cursor A to screen center (SW[7]=0), then move Cursor B to the same spot (SW[7]=1).

Expected:

Both cursors overlap. The 3x3 block sits inside the 10x10 box. Both are cyan.

PART 3 — Grayscale Tests

Test 3.1 — Basic Grayscale

SW9	SW8	SW7	SW6	SW5	SW4	SW3	SW2	SW1	SW0
DN	UP	DN	DN	DN	DN	DN	DN	DN	DN

Expected:

Full colour image converts to black-and-white instantly.
Corner markers still show colour (they bypass the filter).

Test 3.2 — Grayscale + Red OFF

SW9	SW8	SW7	SW6	SW5	SW4	SW3	SW2	SW1	SW0
DN	UP	DN	DN	DN	DN	DN	UP	UP	DN

Expected:

Darker grayscale overall. Red weight (0.2126) is removed.
Faces look darker since skin has high red content.

Test 3.3 — Grayscale + Green OFF

SW9	SW8	SW7	SW6	SW5	SW4	SW3	SW2	SW1	SW0
DN	UP	DN	DN	DN	UP	UP	DN	DN	DN

Expected:

Much darker grayscale. Green has the highest weight (0.7152).
Removing it makes nearly everything very dark.

Test 3.4 — Grayscale + Blue OFF

SW9	SW8	SW7	SW6	SW5	SW4	SW3	SW2	SW1	SW0
DN	UP	DN	UP	UP	DN	DN	DN	DN	DN

Expected:

Slightly darker grayscale. Blue has the lowest weight (0.0722).
Smallest impact of the three channels.

PARTS 1+3 COMBINED — Colour Filter + Grayscale

Test C.1 — Warm Grayscale (Blue removed)

SW9	SW8	SW7	SW6	SW5	SW4	SW3	SW2	SW1	SW0
DN	UP	DN	UP	UP	DN	DN	DN	DN	DN

Expected:

Grayscale calculated without blue. Slightly different tonal response.

Test C.2 — High Contrast Grayscale (Red removed)

SW9	SW8	SW7	SW6	SW5	SW4	SW3	SW2	SW1	SW0
DN	UP	DN	DN	DN	DN	DN	UP	UP	DN

Expected:

Plants/grass appear brighter (more green contribution).

Skin tones appear darker.

RESET Test

Test R.1 — Reset Behaviour

Step 1: Move both cursors to random positions, change some filters.

Step 2: Flip SW[0] UP to reset.

SW9	SW8	SW7	SW6	SW5	SW4	SW3	SW2	SW1	SW0
DN	DN	DN	DN	DN	DN	DN	DN	DN	UP

Step 3: Flip SW[0] DOWN to restart.

SW9	SW8	SW7	SW6	SW5	SW4	SW3	SW2	SW1	SW0
DN	DN	DN	DN	DN	DN	DN	DN	DN	DN

Expected:

Cursor A returns to (100, 100).

Cursor B returns to (300, 200).

Camera image reappears after brief blank.

All filters follow whatever switches are currently set.

Quick Reference Card

What you want to see	Switch Setting
Normal image	All DOWN (SW[0]=0)
Red only	SW[4:3]=11, SW[6:5]=11 (Green+Blue OFF)
Green only	SW[2:1]=11, SW[6:5]=11 (Red+Blue OFF)
Blue only	SW[2:1]=11, SW[4:3]=11 (Red+Green OFF)
Yellow tones	SW[6:5]=11 (Blue OFF)
Cyan tones	SW[2:1]=11 (Red OFF)
Magenta tones	SW[4:3]=11 (Green OFF)
Darker image	SW[2:1]=01, SW[4:3]=01, SW[6:5]=01 (all half)
Very dark image	SW[2:1]=10, SW[4:3]=10, SW[6:5]=10 (all quarter)
Black screen	SW[2:1]=11, SW[4:3]=11, SW[6:5]=11 (all off)
Grayscale	SW[8]=1
Grayscale + no green	SW[8]=1, SW[4:3]=11
Move Cursor A (3x3 block)	SW[7]=0, use KEY[3/2/1/0]
Move Cursor B (10x10 box)	SW[7]=1, use KEY[3/2/1/0]
Reset everything	SW[0] UP then DOWN